

CHANGSHA, China

WMO: 576870

Lat: 28.1095N

Lon: 112.7880E

Elev: 394

StdP: 14.49

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 29.8	(c) 31.8	(d) 16.5	(e) 12.8	(f) 43.1	(g) 21.3	(h) 16.2	(i) 42.7	(j) 17.1	(k) 39.8	(l) 15.5	(m) 41.0	(n) 5.7	(o) 340	(p) 0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	97.5	78.7	95.6	78.7	93.6	78.6	82.0	90.7	81.1	89.8	80.3	88.9	6.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.1	158.3	86.2	79.0	152.8	85.6	78.1	148.0	84.9	46.4	90.9	45.3	89.7	44.4	89.2	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.4	13.2	11.6		DB	27.3	100.4	2.2	1.9	25.7	101.8	24.4	102.9	23.1	103.9	21.5	105.3
				WB	25.7	83.3	2.0	1.4	24.2	84.4	23.0	85.2	21.9	86.0	20.4	87.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	41.9	46.4	54.5	65.2	73.3	79.3	85.5	84.0	76.5	67.0	56.5	46.2
	DBStd	16.15	6.45	8.16	8.33	7.85	5.98	5.24	4.97	5.30	6.39	6.36	7.26	5.74
	HDD50	629	263	153	45	2	0	0	0	0	0	0	23	143
	HDD65	2583	716	522	342	93	6	0	0	0	3	52	267	582
	CDD50	6031	12	52	184	458	723	880	1102	1052	796	527	219	26
	CDD65	2510	0	2	15	99	264	430	637	587	349	114	13	0
	CDH74	26069	1	17	100	660	1958	4159	8359	7138	3027	590	60	0
CDH80	11762	0	2	21	175	651	1659	4442	3476	1193	135	8	0	
Wind	WSAvg	5.2	5.4	5.5	5.3	5.4	5.0	4.7	5.3	5.3	5.3	5.2	5.1	5.3
Precipitation	PrecAvg	58.6	3.0	3.5	5.7	7.2	8.3	9.0	6.0	4.5	3.3	2.6	3.4	2.1
	PrecMax	70.0	6.9	7.6	14.2	10.8	17.6	19.5	14.9	12.2	8.6	6.1	8.2	5.4
	PrecMin	38.5	0.6	0.5	3.3	3.2	2.0	4.8	0.4	1.0	0.1	0.1	0.5	0.0
	PrecStd	8.8	1.5	1.5	2.4	2.2	3.4	3.7	3.9	2.4	2.2	1.4	2.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.4	76.4	82.7	89.0	92.7	95.3	99.9	100.5	96.6	88.6	80.2	66.7
		MCWB	54.3	62.4	69.3	71.4	74.8	78.5	78.8	77.8	77.6	70.6	65.2	54.2
	2%	DB	59.5	68.4	76.2	84.9	89.7	93.1	97.9	97.6	93.1	83.7	74.9	62.1
		MCWB	49.6	57.2	63.6	69.9	73.6	78.5	78.9	78.7	77.7	68.8	62.6	51.0
	5%	DB	55.3	62.9	71.6	81.2	86.8	91.0	96.1	95.3	90.0	80.1	70.9	58.3
		MCWB	47.2	54.0	60.6	67.9	72.4	77.8	78.8	79.2	76.8	66.7	59.8	49.1
	10%	DB	51.3	57.9	67.2	77.5	83.6	88.4	94.3	92.7	86.8	77.0	67.4	54.9
		MCWB	45.4	50.7	58.5	66.8	71.8	77.1	78.7	79.1	75.1	65.8	58.1	47.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.9	63.1	70.4	75.1	78.7	81.4	83.0	83.6	81.4	75.1	68.2	57.8
		MCDB	64.1	74.7	81.7	86.2	87.6	90.8	92.1	92.1	90.2	84.1	76.2	62.4
	2%	WB	51.7	59.0	65.9	72.7	76.7	80.1	81.9	82.4	79.8	71.9	64.8	53.5
		MCDB	56.6	66.7	73.2	81.1	84.4	89.0	91.0	90.7	88.3	78.5	71.6	58.3
	5%	WB	48.3	55.1	62.8	70.6	75.2	79.2	81.0	81.3	78.5	70.0	62.3	51.0
		MCDB	53.4	60.7	69.0	77.7	82.4	87.7	90.1	89.5	86.5	76.8	67.9	55.8
	10%	WB	45.8	51.7	59.7	68.5	73.8	78.3	80.2	80.4	77.2	68.2	59.8	48.8
		MCDB	50.1	57.0	65.2	75.5	80.7	86.2	89.3	88.5	84.6	73.5	65.0	53.6
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.9	11.1	12.3	12.3	11.2	12.0	11.8	12.1	12.0	11.6	10.6
		MCDBR	18.2	19.3	19.4	19.0	17.8	14.7	14.6	15.2	16.3	17.7	18.5	18.9
		MCWBR	9.9	10.1	9.4	7.9	6.3	4.8	3.6	3.8	4.9	5.9	8.0	9.3
	5% WB	MCDBR	15.1	17.0	16.8	16.1	14.0	12.7	13.8	13.9	14.2	14.2	14.3	14.1
MCWBR		9.1	9.9	9.7	8.5	6.7	5.3	4.6	4.9	5.1	6.4	7.8	8.0	
Clear-Sky Solar Irradiance	taub		0.705	0.745	0.790	0.772	0.700	0.683	0.604	0.679	0.710	0.751	0.700	0.681
	taud		1.475	1.441	1.399	1.473	1.633	1.684	1.877	1.681	1.597	1.473	1.534	1.536
	Ebn at Noon		169	178	182	193	209	212	229	210	197	174	169	167
	Edh at Noon		82	92	101	96	82	78	64	77	82	88	77	75
All-Sky Solar Radiation	RadAvg		533	637	852	1066	1218	1306	1674	1444	1195	936	743	611
	RadStd		113	179	95	137	181	156	173	125	131	146	128	136

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) +0.70	(e) N/A	(f) -3.96	(g) N/A	(h) -0.90	(i) -0.92	(j) N/A	(k) N/A	(l) +297	(m) N/A
Regional (6 neighbors)	+0.43	N/A	N/A	+0.79	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page